LAB 3 REPORT

CYOP 300 - BUILDING SECURE WEB APPLICATIONS

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Executive Summary

An observation: I did see the remark about my usage of simple-term-menu, and how it is Unix or Unix-like specific. I had not thought about that when assembling Lab 2, and was well into Lab 3 by the time I saw the remark. I've spent so long working in Unix and Unix-like environments, that I hadn't thought to make sure that the application ran cleanly under Windows. Thank you for taking the time to run the project under WSL.

As with lab 2, I made use of simple-term-menu to create a user interface in the CLI. For this lab, to accommodate the display of the State flowers, I additionally made use of term-image to display jpg files of each flower in the CLI.

For this lab, I created a csv file containing the State data (sourced from https://www.crestcapital.com/tax/us_states_and_capitals), and downloaded the images for each State flower from https://statesymbolsusa.org/categories/flower, naming each file corresponding to the 2-character State abbreviation.

While not discussed in the Lab instructions, I provided a function to save the population data modified in menu item 4, so that the data can persist across multiple executions of the program. Please note that this data is not saved by default – the user is informed and given a choice whether to save the changed data or not. Tests 10-17 demonstrate this by showing a top 5 population graph (test 10), modifying the data for Ohio (test 11), then showing an updated graph with Ohio's population increasing enough to become the most populous state (test 12). Then the application exits, and when executed again, once again shows California as the most populous state (test 14). Then the Ohio population data is modified again, saved (test 15), and, after restarting the program (test 16), when a new graph is displayed, Ohio is once again the most populous State (test 17)

Testing

Test Case Table

Test Cases	Description	Expected Output	Actual Output	Pass/Fail
Test Case 1	Select 1. Display all state information	All 50 states printed, menu printed with Retry/ Main Menu/Exit	All 50 states printed, menu printed with Retry/ Main Menu/Exit	Pass
Test Case 2	Select 1. Display all state information, then select Re-Run this function	All 50 states printed, menu printed with Retry/ Main Menu/Exit, then all 50 states printed, menu printed with Retry/ Main Menu/Exit again	All 50 states printed, menu printed with Retry/ Main Menu/Exit, then all 50 states printed, menu printed with Retry/ Main Menu/Exit again	Pass
Test Case 3	Select 1. Display all state information, then select Exit	All 50 states printed, menu printed with Retry/ Main Menu/Exit, then a goodbye message and the application terminates	All 50 states printed, menu printed with Retry/ Main Menu/Exit, then a goodbye message and the application terminates	Pass
Test Case 4	Select 1. Display all state information, then select Return to Main Menu	All 50 states printed, menu printed with Retry/ Main Menu/Exit, then the top menu is displayed again	All 50 states printed, menu printed with Retry/ Main Menu/Exit, then the top menu is displayed again	Pass
Test Case 5	Select 2. Search, then search by abbrev, enter NV as the state	Image displayed, followed by this text: State: Nevada, Capital: Carson City, Population: 3,104,614, Flower: Sagebrush	Image displayed, followed by this text: State: Nevada, Capital: Carson City, Population: 3,104,614, Flower: Sagebrush	Pass
Test Case 6	Select 2. Search, then search by abbrev, enter no as the state	Error message is displayed, followed by a menu: Invalid state abbreviation. Would you like to try again, return to the main menu, or exit? > Re-run this function Return to Main Menu Exit	Error message is displayed, followed by a menu: Invalid state abbreviation. Would you like to try again, return to the main menu, or exit? > Re-run this function Return to Main Menu Exit	Pass
Test Case 7	Select 2. Search, then search by name, enter iOWa as the state	Image displayed, followed by this text: State: Iowa, Capital: Des Moines, Population: 3,190,369, Flower: Wild Prairie Rose	Image displayed, followed by this text: State: Iowa, Capital: Des Moines, Population: 3,190,369, Flower: Wild Prairie Rose	Pass

Test Case 8	Select 2. Search, then search by name, enter Alabaster as the state	Error message is displayed, followed by a menu: Invalid state name. Would you like to try again, return to the main menu, or exit? > Re-run this function Return to Main Menu Exit	Error message is displayed, followed by a menu: Invalid state name. Would you like to try again, return to the main menu, or exit? > Re-run this function Return to Main Menu Exit	Pass
Test Case 9	Select 2. Search, then choose from menu, select Alabama	Image is displayed, followed by this text: State: Alabama, Capital: Montgomery, Population: 5,024,279, Flower: Camellia	Image is displayed, followed by this text: State: Alabama, Capital: Montgomery, Population: 5,024,279, Flower: Camellia	Pass
Test Case 10	Select 3. Graph	Graph displayed containing the following 5 states and populations: California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187 New York, 29,201,249 Pennsylvania, 13,002,700	Graph displayed containing the following 5 states and populations: California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187 New York, 29,201,249 Pennsylvania, 13,002,700	Pass
Test Case 11	Select 4. Update; search state by abbrev; enter oH, enter new population of 123456789; select N for save changes	Updating the overall state population for a specific state. Select the State you want to update: Enter the state abbreviation: oH Current population of Ohio: 11,799,448 Enter the new population: 123456789 New population of Ohio: 123,456,789 N.b., you must save the changes in order to persist them. They will otherwise be discarded when leaving this program. Save changes? (y/n): n	Updating the overall state population for a specific state. Select the State you want to update: Enter the state abbreviation: oH Current population of Ohio: 11,799,448 Enter the new population: 123456789 New population of Ohio: 123,456,789 N.b., you must save the changes in order to persist them. They will otherwise be discarded when leaving this program. Save changes? (y/n): n	Pass
Test Case 12	Select 3. Graph	Graph displayed containing the following 5 states and populations: Ohio, 123,456,789	Graph displayed containing the following 5 states and populations: Ohio, 123,456,789	Pass

		California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187 New York, 29,201,249	California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187 New York, 29,201,249	
Test Case 13	Select 5. Exit	Application displays a goodbye message and the application terminates	Application displays a goodbye message and the application terminates	Pass
Test Case 14	Select 3. Graph	Graph displayed containing the following 5 states and populations: California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187 New York, 29,201,249 Pennsylvania, 13,002,700	Graph displayed containing the following 5 states and populations: California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187 New York, 29,201,249 Pennsylvania, 13,002,700	Pass
Test Case 15	Select 4. Update; search state by abbrev; enter oH, enter new population of 123,456,789; select y for save changes	Updating the overall state population for a specific state. Select the State you want to update: Enter the state abbreviation: oH Current population of Ohio: 11,799,448 Enter the new population: 123,456,789 New population of Ohio: 123,456,789 N.b., you must save the changes in order to persist them. They will otherwise be discarded when leaving this program. Save changes? (y/n): y	Updating the overall state population for a specific state. Select the State you want to update: Enter the state abbreviation: oH Current population of Ohio: 11,799,448 Enter the new population: 123,456,789 New population of Ohio: 123,456,789 N.b., you must save the changes in order to persist them. They will otherwise be discarded when leaving this program. Save changes? (y/n): y	Pass
Test Case 16	Select 5. Exit	Application displays a goodbye message and the application terminates	Application displays a goodbye message and the application terminates	Pass
Test Case 17	Select 3. Graph	Graph displayed containing the following 5 states and populations: Ohio, 123,456,789 California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187	Graph displayed containing the following 5 states and populations: Ohio, 123,456,789 California, 39,538,223 Texas, 29,145,505 Florida, 21,538,187	Pass

		New York, 29,201,249	New York, 29,201,249	

Test Case Screenshots

Test Case 1

```
Displaying all U.S. States in Alphabetical order along with the Capital, State Population, and Flower.
State: Alabama, Capital: Montgomery, Population: 5,024,279, Flower: Camellia
State: Alaska, Capital: Juneau, Population: 733,391, Flower: Forget-Me-Not
State: Arizona, Capital: Phoenix, Population: 7,151,502, Flower: Saguaro Cactus Blossom
State: Arkansas, Capital: Little Rock, Population: 3,011,524, Flower: Apple Blossom
State: California, Capital: Sacramento, Population: 39,538,223, Flower: Golden Poppy
State: Colorado, Capital: Denver, Population: 5,773,714, Flower: Rocky Mountain Columbine
State: Connecticut, Capital: Hartford, Population: 3,605,944, Flower: Mountain Laurel
State: Delaware, Capital: Dover, Population: 989,948, Flower: Peach Blossom
State: Florida, Capital: Tallahassee, Population: 21,538,187, Flower: Orange Blossom
State: Georgia, Capital: Atlanta, Population: 10,711,908, Flower: Cherokee Rose
State: Hawaii, Capital: Honolulu, Population: 1,455,271, Flower: Yellow Hibiscus
State: Idaho, Capital: Boise, Population: 1,839,106, Flower: Syringa
State: Illinois, Capital: Springfield, Population: 12,812,508, Flower: Violet
State: Indiana, Capital: Indianapolis, Population: 6,785,528, Flower: Peony
State: Iowa, Capital: Des Moines, Population: 3,190,369, Flower: Wild Prairie Rose
State: Kansas, Capital: Topeka, Population: 2,937,880, Flower: Sunflower
State: Kentucky, Capital: Frankfort, Population: 4,505,836, Flower: Goldenrod
State: Louisiana, Capital: Baton Rouge, Population: 4,657,757, Flower: Magnolia
State: Maine, Capital: Augusta, Population: 1,362,359, Flower: White Pine Cone and Tassel
State: Maryland, Capital: Annapolis, Population: 6,177,224, Flower: Black-Eyed Susan
State: Massachusetts, Capital: Boston, Population: 7,029,917, Flower: Mayflower
State: Michigan, Capital: Lansing, Population: 10,077,331, Flower: Apple Blossom
State: Minnesota, Capital: Saint Paul, Population: 5,706,494, Flower: Pink and White Lady's Slipper
State: Mississippi, Capital: Jackson, Population: 2,961,279, Flower: Magnolia
State: Missouri, Capital: Jefferson City, Population: 6,154,913, Flower: White Hawthorn Blossom
State: Montana, Capital: Helena, Population: 1,084,225, Flower: Bitterroot
State: Nebraska, Capital: Lincoln, Population: 1,961,504, Flower: Goldenrod
State: Nevada, Capital: Carson City, Population: 3,104,614, Flower: Sagebrush
State: New Hampshire, Capital: Concord, Population: 1,377,529, Flower: Purple Lilac
State: New Jersey, Capital: Trenton, Population: 9,288,994, Flower: Common Blue Violet
State: New Mexico, Capital: Santa Fe, Population: 2,117,522, Flower: Yucca Flower
State: New York, Capital: Albany, Population: 20,201,249, Flower: Rose
State: North Carolina, Capital: Raleigh, Population: 10,439,388, Flower: Dogwood
State: North Dakota, Capital: Bismarck, Population: 779,094, Flower: Wild Prairie Rose
State: Ohio, Capital: Columbus, Population: 11,799,448, Flower: Scarlet Carnation
State: Oklahoma, Capital: Oklahoma City, Population: 3,959,353, Flower: Mistletoe
State: Oregon, Capital: Salem, Population: 4,237,256, Flower: Oregon Grape
State: Pennsylvania, Capital: Harrisburg, Population: 13,002,700, Flower: Mountain Laurel
State: Rhode Island, Capital: Providence, Population: 1,097,379, Flower: Violet
State: South Carolina, Capital: Columbia, Population: 5,118,425, Flower: Yellow Jessamine
State: South Dakota, Capital: Pierre, Population: 886,667, Flower: Pasque Flower
State: Tennessee, Capital: Nashville, Population: 6,910,840, Flower: Iris
State: Texas, Capital: Austin, Population: 29,145,505, Flower: Bluebonnet
State: Utah, Capital: Salt Lake City, Population: 3,271,616, Flower: Sego Lily
State: Vermont, Capital: Montpelier, Population: 643,077, Flower: Red Clover
State: Virginia, Capital: Richmond, Population: 8,631,393, Flower: American Dogwood
State: Washington, Capital: Olympia, Population: 7,705,281, Flower: Coast Rhododendron
State: West Virginia, Capital: Charleston, Population: 1,793,716, Flower: Rhododendron
State: Wisconsin, Capital: Madison, Population: 5,893,718, Flower: Wood Violet
State: Wyoming, Capital: Cheyenne, Population: 576,851, Flower: Indian Paintbrush
> Re-run this function
  Return to Main Menu
  Exit
```

Test Case 2

```
State: West Virginia, Capital: Charleston, Population: 1,793,716, Flower: Rhododendron
State: Wisconsin, Capital: Madison, Population: 5,893,718, Flower: Wood Violet
State: Wyoming, Capital: Cheyenne, Population: 576,851, Flower: Indian Paintbrush
Displaying all U.S. States in Alphabetical order along with the Capital, State Population, and Flower.
State: Alabama, Capital: Montgomery, Population: 5,024,279, Flower: Camellia
State: Alaska, Capital: Juneau, Population: 733,391, Flower: Forget-Me-Not
State: Arizona, Capital: Phoenix, Population: 7,151,502, Flower: Saguaro Cactus Blossom
State: Arkansas, Capital: Little Rock, Population: 3,011,524, Flower: Apple Blossom
State: California, Capital: Sacramento, Population: 39,538,223, Flower: Golden Poppy
State: Colorado, Capital: Denver, Population: 5,773,714, Flower: Rocky Mountain Columbine
State: Connecticut, Capital: Hartford, Population: 3,605,944, Flower: Mountain Laurel
State: Delaware, Capital: Dover, Population: 989,948, Flower: Peach Blossom
State: Florida, Capital: Tallahassee, Population: 21,538,187, Flower: Orange Blossom
State: Georgia, Capital: Atlanta, Population: 10,711,908, Flower: Cherokee Rose
State: Hawaii, Capital: Honolulu, Population: 1,455,271, Flower: Yellow Hibiscus
State: Idaho, Capital: Boise, Population: 1,839,106, Flower: Syringa
State: Illinois, Capital: Springfield, Population: 12,812,508, Flower: Violet
State: Indiana, Capital: Indianapolis, Population: 6,785,528, Flower: Peony
State: Iowa, Capital: Des Moines, Population: 3,190,369, Flower: Wild Prairie Rose
State: Kansas, Capital: Topeka, Population: 2,937,880, Flower: Sunflower
State: Kentucky, Capital: Frankfort, Population: 4,505,836, Flower: Goldenrod
State: Louisiana, Capital: Baton Rouge, Population: 4,657,757, Flower: Magnolia
State: Maine, Capital: Augusta, Population: 1,362,359, Flower: White Pine Cone and Tassel
State: Maryland, Capital: Annapolis, Population: 6,177,224, Flower: Black-Eyed Susan
State: Massachusetts, Capital: Boston, Population: 7,029,917, Flower: Mayflower
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State: Minnesota, Capital: Saint Paul, Population: 5,706,494, Flower: Pink and White Lady's Slipper
State: Mississippi, Capital: Jackson, Population: 2,961,279, Flower: Magnolia
State: Missouri, Capital: Jefferson City, Population: 6,154,913, Flower: White Hawthorn Blossom
State: Montana, Capital: Helena, Population: 1,084,225, Flower: Bitterroot
State: Nebraska, Capital: Lincoln, Population: 1,961,504, Flower: Goldenrod
State: Nevada, Capital: Carson City, Population: 3,104,614, Flower: Sagebrush
State: New Hampshire, Capital: Concord, Population: 1,377,529, Flower: Purple Lilac
State: New Jersey, Capital: Trenton, Population: 9,288,994, Flower: Common Blue Violet
State: New Mexico, Capital: Santa Fe, Population: 2,117,522, Flower: Yucca Flower
State: New York, Capital: Albany, Population: 20,201,249, Flower: Rose
State: North Carolina, Capital: Raleigh, Population: 10,439,388, Flower: Dogwood
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State: Rhode Island, Capital: Providence, Population: 1,097,379, Flower: Violet
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State: Texas, Capital: Austin, Population: 29,145,505, Flower: Bluebonnet
State: Utah, Capital: Salt Lake City, Population: 3,271,616, Flower: Sego Lily
State: Vermont, Capital: Montpelier, Population: 643,077, Flower: Red Clover
State: Virginia, Capital: Richmond, Population: 8,631,393, Flower: American Dogwood
State: Washington, Capital: Olympia, Population: 7,705,281, Flower: Coast Rhododendron
State: West Virginia, Capital: Charleston, Population: 1,793,716, Flower: Rhododendron
State: Wisconsin, Capital: Madison, Population: 5,893,718, Flower: Wood Violet
State: Wyoming, Capital: Cheyenne, Population: 576,851, Flower: Indian Paintbrush
> Re-run this function
  Return to Main Menu
  Exit
```

Test Case 3

```
State: Wisconsin, Capital: Madison, Population: 5,893,718, Flower: Wood Violet
State: Wyoming, Capital: Cheyenne, Population: 576,851, Flower: Indian Paintbrush
Displaying all U.S. States in Alphabetical order along with the Capital, State Population, and Flower.
State: Alabama, Capital: Montgomery, Population: 5,024,279, Flower: Camellia
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State: Arkansas, Capital: Little Rock, Population: 3,011,524, Flower: Apple Blossom
State: California, Capital: Sacramento, Population: 39,538,223, Flower: Golden Poppy
State: Colorado, Capital: Denver, Population: 5,773,714, Flower: Rocky Mountain Columbine
State: Connecticut, Capital: Hartford, Population: 3,605,944, Flower: Mountain Laurel
State: Delaware, Capital: Dover, Population: 989,948, Flower: Peach Blossom
State: Florida, Capital: Tallahassee, Population: 21,538,187, Flower: Orange Blossom
State: Georgia, Capital: Atlanta, Population: 10,711,908, Flower: Cherokee Rose
State: Hawaii, Capital: Honolulu, Population: 1,455,271, Flower: Yellow Hibiscus
State: Idaho, Capital: Boise, Population: 1,839,106, Flower: Syringa
State: Illinois, Capital: Springfield, Population: 12,812,508, Flower: Violet
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State: Iowa, Capital: Des Moines, Population: 3,190,369, Flower: Wild Prairie Rose
State: Kansas, Capital: Topeka, Population: 2,937,880, Flower: Sunflower
State: Kentucky, Capital: Frankfort, Population: 4,505,836, Flower: Goldenrod
State: Louisiana, Capital: Baton Rouge, Population: 4,657,757, Flower: Magnolia
State: Maine, Capital: Augusta, Population: 1,362,359, Flower: White Pine Cone and Tassel
State: Maryland, Capital: Annapolis, Population: 6,177,224, Flower: Black-Eyed Susan
State: Massachusetts, Capital: Boston, Population: 7,029,917, Flower: Mayflower
State: Michigan, Capital: Lansing, Population: 10,077,331, Flower: Apple Blossom
State: Minnesota, Capital: Saint Paul, Population: 5,706,494, Flower: Pink and White Lady's Slipper
State: Mississippi, Capital: Jackson, Population: 2,961,279, Flower: Magnolia
State: Missouri, Capital: Jefferson City, Population: 6,154,913, Flower: White Hawthorn Blossom
State: Montana, Capital: Helena, Population: 1,084,225, Flower: Bitterroot
State: Nebraska, Capital: Lincoln, Population: 1,961,504, Flower: Goldenrod
State: Nevada, Capital: Carson City, Population: 3,104,614, Flower: Sagebrush
State: New Hampshire, Capital: Concord, Population: 1,377,529, Flower: Purple Lilac
State: New Jersey, Capital: Trenton, Population: 9,288,994, Flower: Common Blue Violet
State: New Mexico, Capital: Santa Fe, Population: 2,117,522, Flower: Yucca Flower
State: New York, Capital: Albany, Population: 20,201,249, Flower: Rose
State: North Carolina, Capital: Raleigh, Population: 10,439,388, Flower: Dogwood
State: North Dakota, Capital: Bismarck, Population: 779,094, Flower: Wild Prairie Rose
State: Ohio, Capital: Columbus, Population: 11,799,448, Flower: Scarlet Carnation
State: Oklahoma, Capital: Oklahoma City, Population: 3,959,353, Flower: Mistletoe
State: Oregon, Capital: Salem, Population: 4,237,256, Flower: Oregon Grape
State: Pennsylvania, Capital: Harrisburg, Population: 13,002,700, Flower: Mountain Laurel
State: Rhode Island, Capital: Providence, Population: 1,097,379, Flower: Violet
State: South Carolina, Capital: Columbia, Population: 5,118,425, Flower: Yellow Jessamine
State: South Dakota, Capital: Pierre, Population: 886,667, Flower: Pasque Flower
State: Tennessee, Capital: Nashville, Population: 6,910,840, Flower: Iris
State: Texas, Capital: Austin, Population: 29,145,505, Flower: Bluebonnet
State: Utah, Capital: Salt Lake City, Population: 3,271,616, Flower: Sego Lily
State: Vermont, Capital: Montpelier, Population: 643,077, Flower: Red Clover
State: Virginia, Capital: Richmond, Population: 8,631,393, Flower: American Dogwood
State: Washington, Capital: Olympia, Population: 7,705,281, Flower: Coast Rhododendron
State: West Virginia, Capital: Charleston, Population: 1,793,716, Flower: Rhododendron
State: Wisconsin, Capital: Madison, Population: 5,893,718, Flower: Wood Violet
State: Wyoming, Capital: Cheyenne, Population: 576,851, Flower: Indian Paintbrush
Exiting program. Thank you for using me!
```
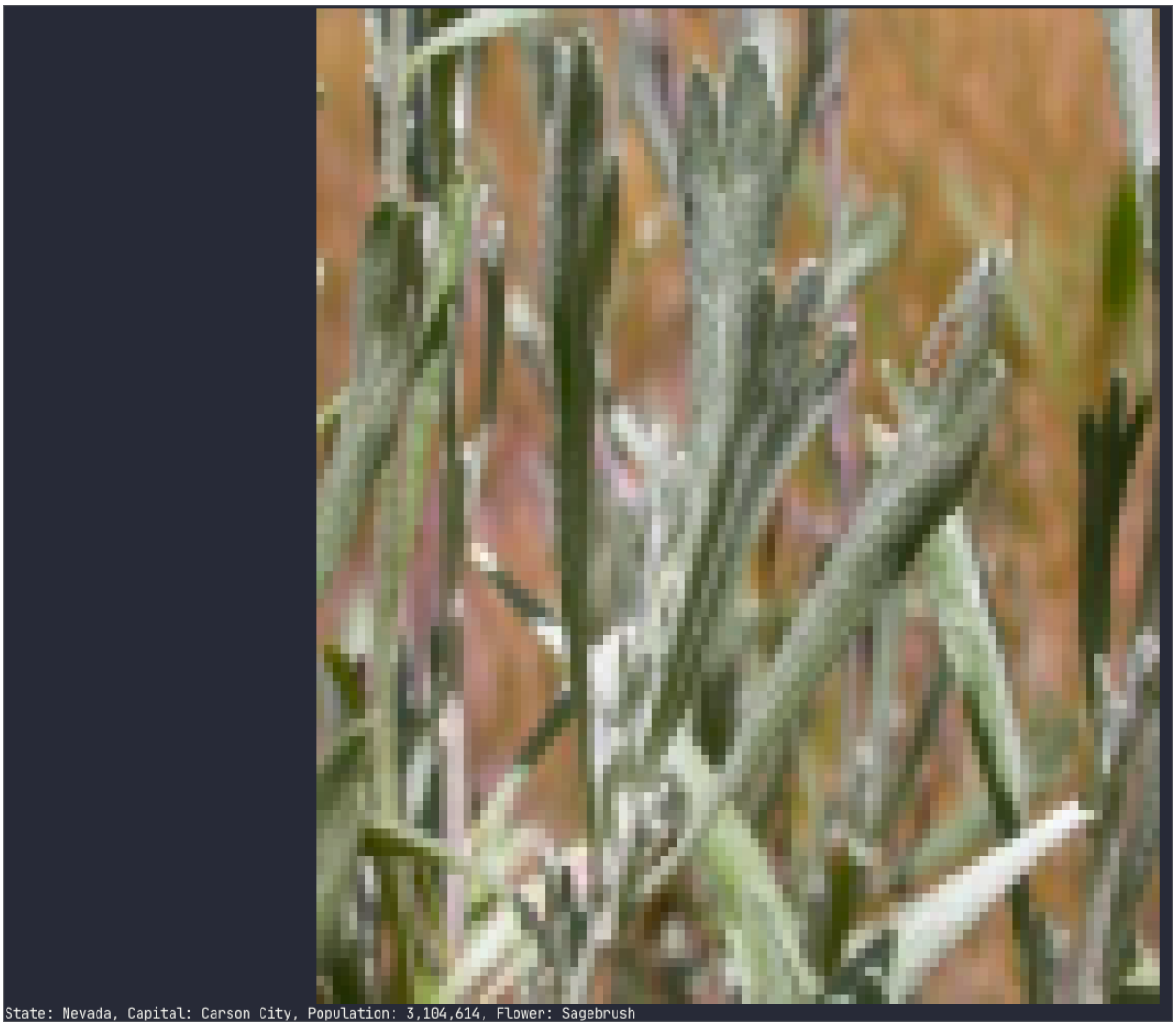

Test Case 4

```
Welcome to Lab 3. Please select from one of the following options:
> 1. Display all U.S. States in Alphabetical order along with the Capital, State Population, and Flower
  2. Search for a specific state and display the appropriate Capital name, State Population, and an image of the associated State Flower.
  3. Provide a Bar graph of the top 5 populated States showing their overall population.
  4. Update the overall state population for a specific state.
  5. Exit the program
```

Test Case 5

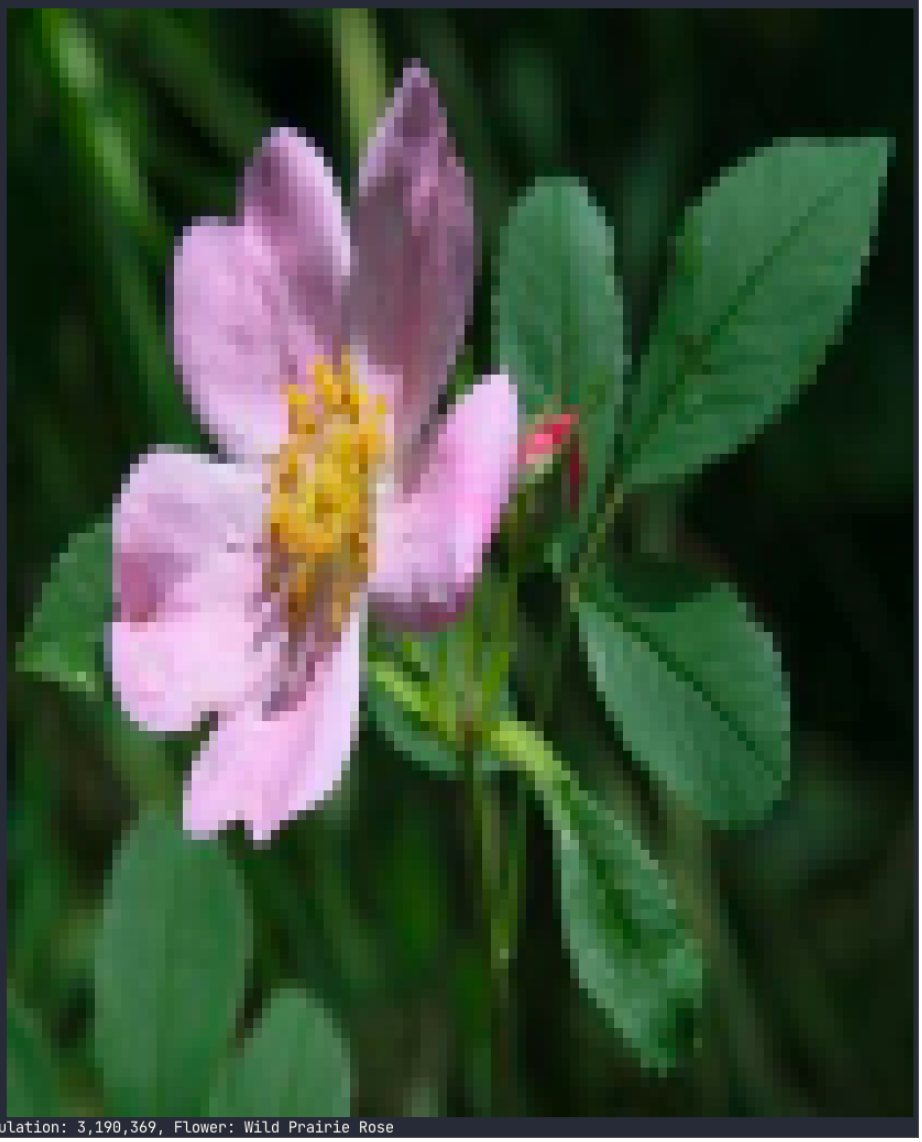


Test Case 6

```
Searching for a specific state and displaying the appropriate Capital name, State Population, and an image of the associated State Flower.
Enter the state abbreviation: no
Invalid state abbreviation.
Would you like to try again, return to the main menu, or exit?
```

Test Case 7

```
Searching for a specific state and displaying the appropriate Capital name, State Population, and an image of the associated State Flower.
Enter the state name: iOWa
```

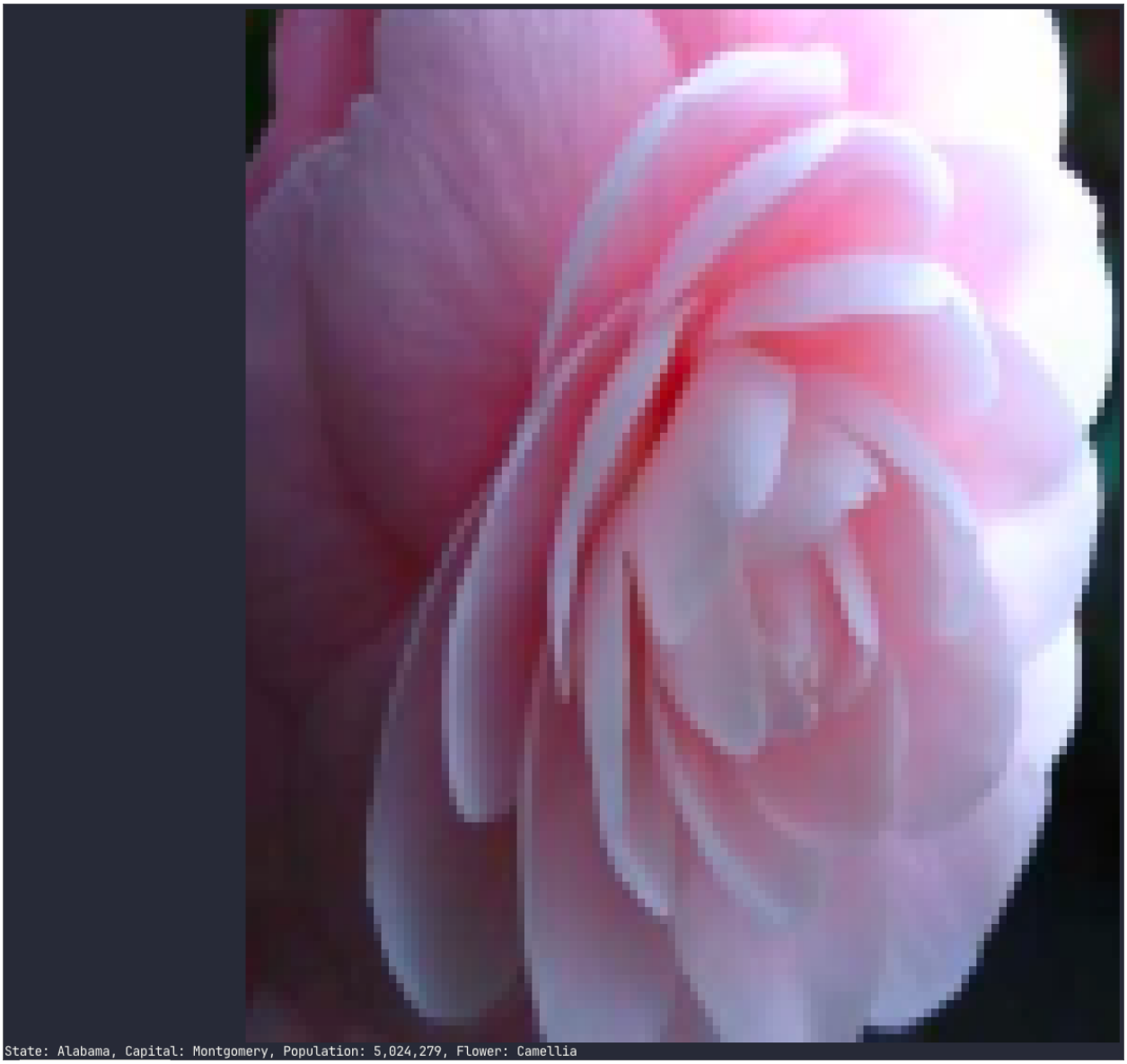


```
State: Iowa, Capital: Des Moines, Population: 3,190,369, Flower: Wild Prairie Rose
```

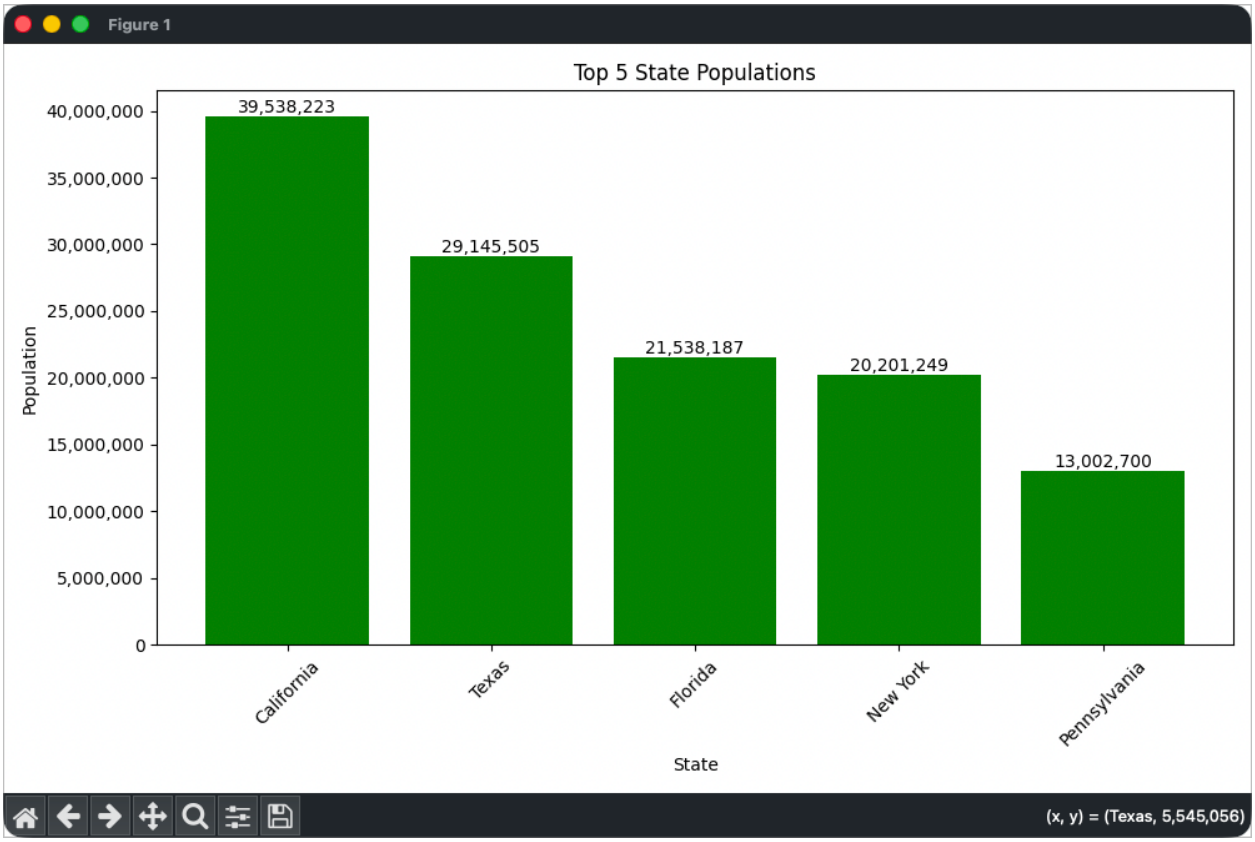
Test Case 8

```
Enter the state name: Alabaster
Invalid state name.
Would you like to try again, return to the main menu, or exit?
> Re-run this function
  Return to Main Menu
  Exit
```

Test Case 9


State: Alabama, Capital: Montgomery, Population: 5,024,279, Flower: Camellia

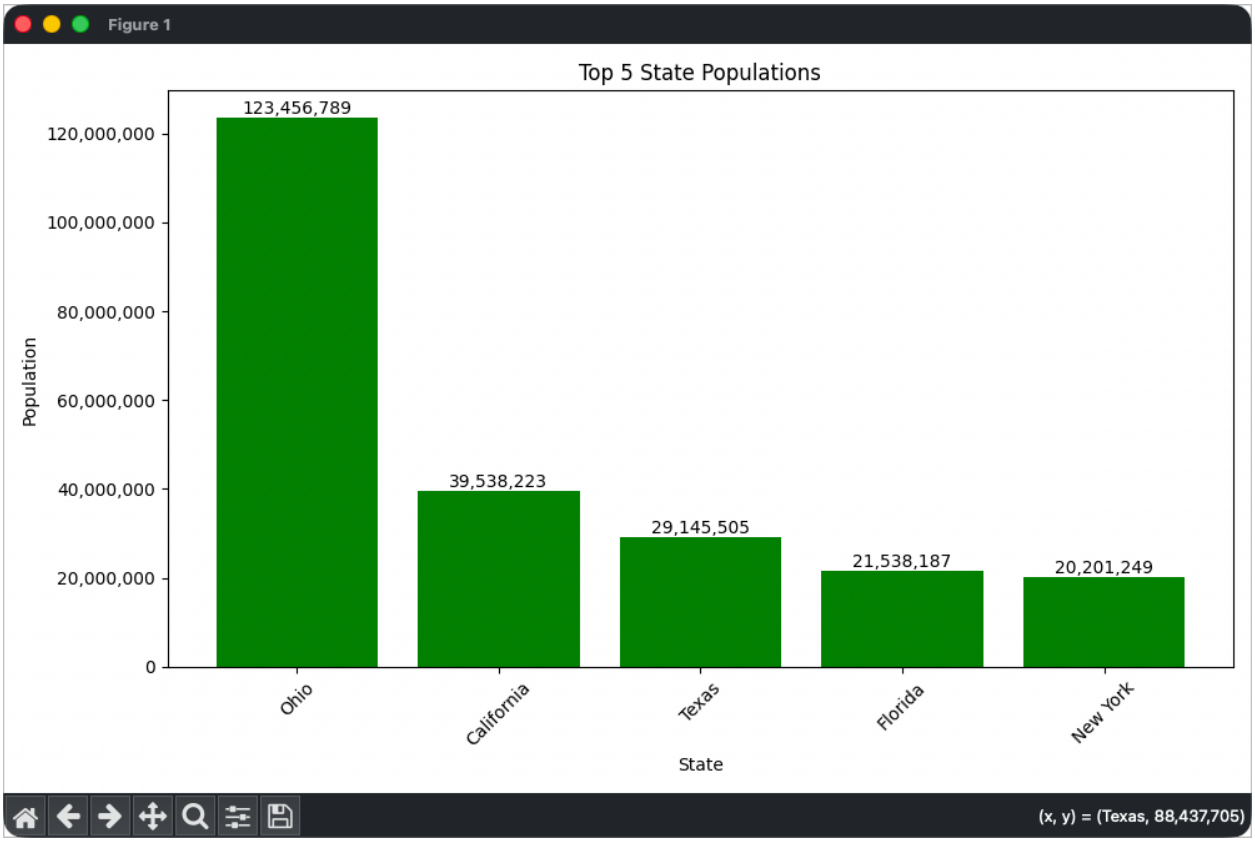
Test Case 10



Test Case 11

```
Welcome to Lab 3. Please select from one of the following options:
Updating the overall state population for a specific state.
Select the State you want to update:
Enter the state abbreviation: oH
Current population of Ohio: 11,799,448
Enter the new population: 123456789
New population of Ohio: 123,456,789
N.b., you must save the changes in order to persist them.
They will otherwise be discarded when leaving this program.
Save changes? (y/n): n
> Re-run this function
  Return to Main Menu
  Exit
```

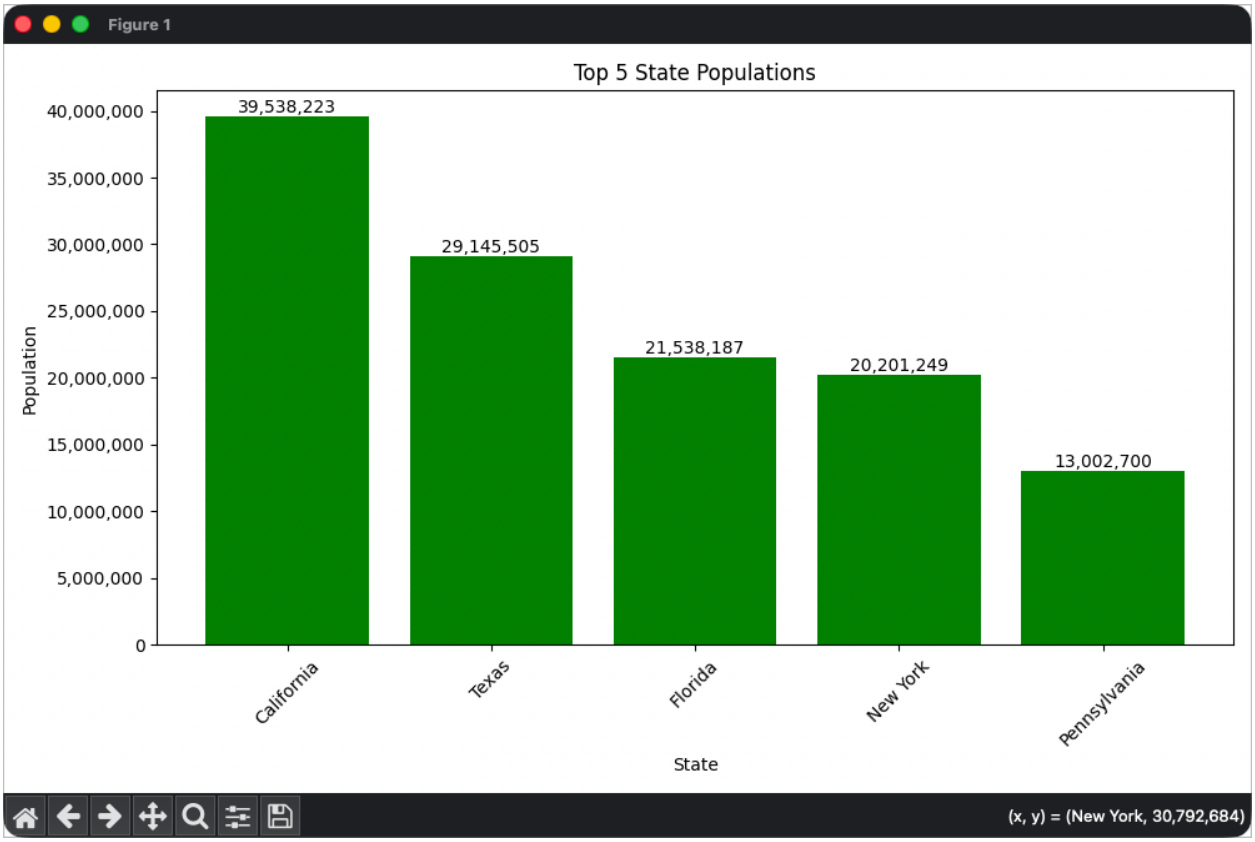
Test Case 12



Test Case 13

```
Welcome to Lab 3. Please select from one of the following options:
Exiting program. Thank you for using me!
```

Test Case 14



Test Case 15

```
Welcome to Lab 3. Please select from one of the following options:
Updating the overall state population for a specific state.
Select the State you want to update:
Enter the state abbreviation: oH
Current population of Ohio: 11,799,448
Enter the new population: 123,456,789
New population of Ohio: 123,456,789
N.b., you must save the changes in order to persist them.
They will otherwise be discarded when leaving this program.
Save changes? (y/n): y
```

Test Case 16

```
Welcome to Lab 3. Please select from one of the following options:
Exiting program. Thank you for using me!
```

Test Case 17

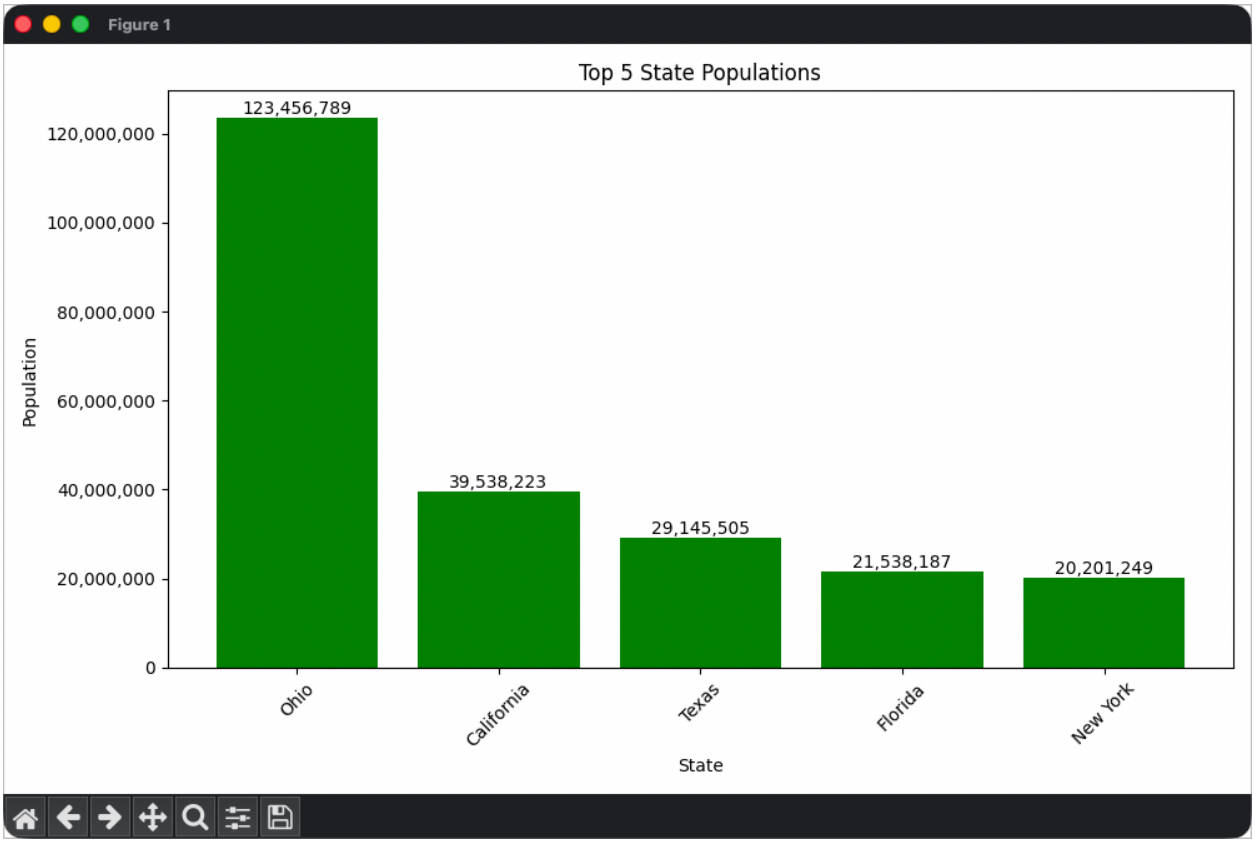


Figure 1
Top 5 State Populations
123,456,789
39,538,223
29,145,505
21,538,187
20,201,249
120,000,000
100,000,000
80,000,000
60,000,000
40,000,000
20,000,000
0
Population
Ohio
California
Texas
Florida
New York
State

Pylint Results

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Run 1:
λ pylint *.py
************* Module lab3.lab3graph
lab3graph.py:28:0: C0301: Line too long (102/100) (line-too-long)
lab3graph.py:51:8: C0104: Disallowed name "bar" (disallowed-name)
lab3graph.py:38:4: W0612: Unused variable 'figure' (unused-variable)
************* Module lab3.lab3main
lab3main.py:50:11: W0108: Lambda may not be necessary (unnecessary-lambda)
lab3main.py:88:11: W0108: Lambda may not be necessary (unnecessary-lambda)
lab3main.py:111:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
lab3main.py:145:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
lab3main.py:184:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
************* Module lab3.lab3prompt
lab3prompt.py:73:0: C0301: Line too long (106/100) (line-too-long)
lab3prompt.py:71:4: E0213: Method 'indexed_menu' should have "self" as
first argument (no-self-argument)
lab3prompt.py:16:0: C0411: standard import "typing.Any" should be placed
before third party import "simple_term_menu.TerminalMenu" (wrong-import-
order)
************* Module lab3.lab3search_states
lab3search_states.py:26:0: W0702: No exception type(s) specified (bare-
except)
lab3search_states.py:204:0: R1711: Useless return at end of function or
method (useless-return)
lab3search_states.py:221:0: R1711: Useless return at end of function or
method (useless-return)
lab3search_states.py:238:0: R1711: Useless return at end of function or
method (useless-return)
lab3search_states.py:19:0: C0411: standard import "os" should be placed
before third party import "term_image.image.from_file" (wrong-import-
order)
lab3search_states.py:20:0: C0411: standard import "typing.Optional" should
be placed before third party import "term_image.image.from_file" (wrong-
import-order)
lab3search_states.py:25:4: W0611: Unused exit_program imported from
lab3.lab3common (unused-import)
lab3search_states.py:25:4: W0611: Unused rerun_or_return imported from
lab3.lab3common (unused-import)
************* Module lab3.lab3states
lab3states.py:85:13: W1514: Using open without explicitly specifying an
encoding (unspecified-encoding)
lab3states.py:104:13: W1514: Using open without explicitly specifying an
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encoding (unspecified-encoding)
************* Module lab3.main
main.py:27:0: C0116: Missing function or method docstring (missing-
function-docstring)

------------------------------------------------------------------
Your code has been rated at 9.02/10

Run 2:
λ pylint *.py
************* Module lab3.lab3main
lab3main.py:50:11: W0108: Lambda may not be necessary (unnecessary-lambda)
lab3main.py:88:11: W0108: Lambda may not be necessary (unnecessary-lambda)
lab3main.py:111:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
lab3main.py:145:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
lab3main.py:184:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
************* Module lab3.lab3prompt
lab3prompt.py:73:0: C0301: Line too long (106/100) (line-too-long)
lab3prompt.py:71:4: E0213: Method 'indexed_menu' should have "self" as
first argument (no-self-argument)
lab3prompt.py:16:0: C0411: standard import "typing.Any" should be placed
before third party import "simple_term_menu.TerminalMenu" (wrong-import-
order)
************* Module lab3.lab3search_states
lab3search_states.py:26:0: W0702: No exception type(s) specified (bare-
except)
lab3search_states.py:204:0: R1711: Useless return at end of function or
method (useless-return)
lab3search_states.py:221:0: R1711: Useless return at end of function or
method (useless-return)
lab3search_states.py:238:0: R1711: Useless return at end of function or
method (useless-return)
lab3search_states.py:19:0: C0411: standard import "os" should be placed
before third party import "term_image.image.from_file" (wrong-import-
order)
lab3search_states.py:20:0: C0411: standard import "typing.Optional" should
be placed before third party import "term_image.image.from_file" (wrong-
import-order)
lab3search_states.py:25:4: W0611: Unused exit_program imported from
lab3.lab3common (unused-import)
lab3search_states.py:25:4: W0611: Unused rerun_or_return imported from
lab3.lab3common (unused-import)
************* Module lab3.lab3states
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lab3states.py:85:13: W1514: Using open without explicitly specifying an
encoding (unspecified-encoding)
lab3states.py:104:13: W1514: Using open without explicitly specifying an
encoding (unspecified-encoding)
************* Module lab3.main
main.py:27:0: C0116: Missing function or method docstring (missing-
function-docstring)

------------------------------------------------------------------
Your code has been rated at 9.14/10 (previous run: 9.02/10, +0.12)

Run 3:
λ pylint *.py
************* Module lab3.lab3main
lab3main.py:50:11: W0108: Lambda may not be necessary (unnecessary-lambda)
lab3main.py:88:11: W0108: Lambda may not be necessary (unnecessary-lambda)
lab3main.py:111:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
lab3main.py:145:11: W0108: Lambda may not be necessary (unnecessary-
lambda)
lab3main.py:184:11: W0108: Lambda may not be necessary (unnecessary-
lambda)

------------------------------------------------------------------
Your code has been rated at 9.81/10 (previous run: 9.14/10, +0.67)
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